Multivariate Calculus Lecture Notes
Section 13.3 – Arc Length and Speed

I. Arc length
   a. Recall the formula for arc length in 2D.
      i. Suppose a path can be described parametrically via $\mathbf{r}(t) = \langle x(t), y(t) \rangle$.
      ii. Then the arc length from $t = a$ to $t = b$ is given by
$$\int_a^b \sqrt{x'(t)^2 + y'(t)^2}\, dt.$$
   b. Suppose $\mathbf{r}(t)$ is differentiable and that $\mathbf{r}'(t)$ is continuous on $[a, b]$. Then the length of the path $\mathbf{r}(t)$ for $a \leq t \leq b$ is given by
$$\int_a^b ||\mathbf{r}'(t)||\, dt = \int_a^b \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}\, dt.$$
   c. Example: Find the arc length of the helix given by $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + t\mathbf{k}$ over one complete loop of the helix.
      i. One complete loop would start, for instance, at $t = 0$ and end at $t = 2\pi$.
      ii. $\mathbf{r}'(t) = -\sin t + \cos t + 1$.
      iii. Thus $||\mathbf{r}'(t)|| = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{2}$.
      iv. Therefore, $s = \displaystyle\int_0^{2\pi} \sqrt{2}\, dt = 2\sqrt{2}\pi$.

II. Speed
   a. We can turn arc length into a function $s(t)$ by defining $s(t)$ as the length along the curve starting at time $t = a$.
      i. Graphically:

      ii. Mathematically: $s(t) = \displaystyle\int_a^t ||\mathbf{r}'(u)||\, du$.
      iii. This function $s(t)$, then, is calculating the distance traveled by an object along the curve $\mathbf{r}(t)$ starting at time $t = a$.
   b. This means that the *speed* of the object at time t is given by
$$\frac{ds}{dt} = \frac{d}{dt}\int_a^t ||\mathbf{r}'(u)||\, du = ||\mathbf{r}'(t)||.$$
   c. This should make sense:
      i. If $\mathbf{r}(t)$ denotes the position vector, then $\mathbf{r}'(t)$ denotes the velocity vector.
      ii. We know that the direction of $\mathbf{r}'(t)$ is the direction the object is moving at time t.
      iii. Now we know that the magnitude of $\mathbf{r}'(t)$ is the speed of the object at time t.
   d. Find the speed at time $t = 2$ of an object traveling along the path $\mathbf{r}(t) = t\mathbf{i} + t^3\mathbf{j} + 3t\mathbf{k}$.
      i. We have $\mathbf{r}'(t) = \mathbf{i} + 3t^2\mathbf{j} + 3\mathbf{k}$.
      ii. Then $\mathbf{r}'(2) = \mathbf{i} + 12\mathbf{j} + 3\mathbf{k}$.
      iii. So the speed is $||\mathbf{r}'(2)|| = \sqrt{1 + 144 + 9} = \sqrt{154}$.

III. Describing a path by arc length
   a. There are infinitely many ways to parameterize a vector-valued function.
   b. We've already seen time as a parameter.
      i. This means that $\mathbf{r}(t)$ tells us where on the path we are after we've traveled t units of time.
      ii. Graphically:

   c. Another way to parameterize a vector-valued function tells us where we are on the path after we've traveled s units of distance.
      i. For example, if we have $\mathbf{r}(s)$, where s represents the distance traveled, then $\mathbf{r}(4)$ tells us where we are on the path after we've traveled 4 distance units.
      ii. Graphically:

      iii. This is called parameterizing by *arc length* because s gives the length (i.e., distance traveled) on the path.
   d. You use the arc length function $s(t)$ to convert from one parametrization to the other.
   e. Example: Parametrize the helix $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + t\mathbf{k}$ in terms of arc length.
      i. First, we need to find the arc length function $s(t)$. We can start with $t = 0$.
      ii. From our calculations above we have $s(t) = \displaystyle\int_0^t \sqrt{2}\, du = \sqrt{2}t$.
      iii. Solving for t, we obtain $t = s/\sqrt{2}$.
      iv. Thus $\mathbf{r}(s) = \cos(s/\sqrt{2})\mathbf{i} + \sin(s/\sqrt{2})\mathbf{j} + (s/\sqrt{2})\mathbf{k}$.
      v. This means that, for example, as s ranges from 0 to 1, $\mathbf{r}(s)$ traces out a path of length 1:

$$\int_0^1 ||\mathbf{r}(s)||\, ds = \int_0^1 \sqrt{\frac{\sin^2(s/\sqrt{2})}{2} + \frac{\cos^2(s/\sqrt{2})}{2} + \frac{1}{2}}\, ds = \int_0^1 \sqrt{\frac{1}{2} + \frac{1}{2}}\, ds = \int_0^1 ds = s\Big|_0^1 = 1.$$

Homework: Section 13.3 #1, 4, 6, 10, 14, 21, 23, 28, 37.